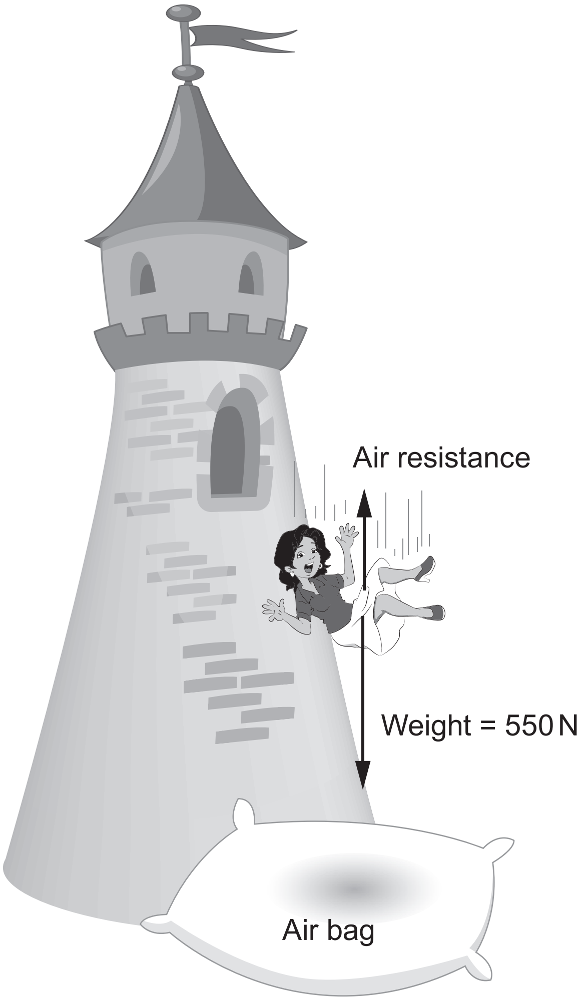
Examiner only

5. As part of a charity event a woman jumps from a tall tower and she safely lands on a large air bag that is beneath.



(a) On Earth, an object of 1 kg has a weight of 10 N. [1]
Calculate the mass of the woman.

Mass = ... kg

Examiner only

(b) The woman hits the air bag with 5.60 kJ of kinetic energy.
The airbag stops her in a distance of 2.8 m.

(i) Convert 5.60 kJ into joules. [1]

Kinetic energy = ... J

(ii) Use the equation:

$$\text{mean force} = \frac{\text{work done}}{\text{distance}}$$

to calculate the mean force bringing the woman to rest. [2]

Mean force = ... N

(iii) It is suggested that it would be safer if the air bag were filled with water. The distance to stop the same woman falling from the same tower would be 0.8 m instead of 2.8 m. Without further calculation explain whether you agree with the suggestion. [3]

....

....

....

....

....

....

7